

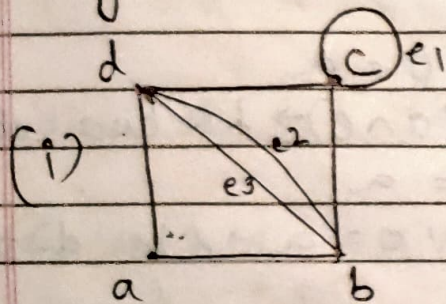
UNIT: 4

Graph Theory

1) A graph is an ordered pair (V, E) of two sets where V is set of vertices, it should be finite & non-empty & E is set of edges & each $e \in E$ corresponds a unique unordered pair $V_1, V_2 \in V$.

2) Degree of Vertex :-

It is a no. which indicates no. of edges incident to that vertex.



$$d(a) = 2$$

$$d(b) = 4$$

$$d(d) = 4$$

$$d(c) = 4$$

3) Parallel edges :-

If two edges have same starting & ending vertex, then they are called parallel edges or multiple edges.

4) Loop :- If an edge starts and ends in the same vertex, then it is called a loop.

In above graph (i), e_1 is a loop
 e_2, e_3 are parallel edges.

* Types of Graph:-

1) Multigraph:-

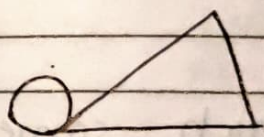
A graph in which a pair of vertices is join by two or more edges is called multigraph.

2) Pseudograph:-

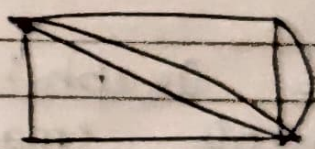
A graph having loops but not multiple edges is called pseudograph.

3) Simple graph

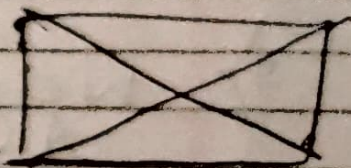
A graph without loops and multiple edges is called a simple graph.



Pseudograph



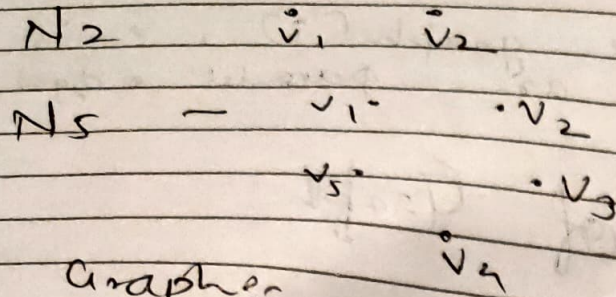
Multi-g



Simple-g

4) Null graph:-

A graph in which set of edges is empty is called Null graph.
Null graph on n vertices is denoted by N_n .

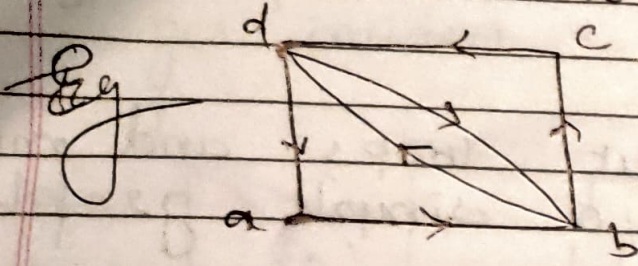


5) Finite Graph:-

A graph which has finite no. of vertices & edges is called finite graph.

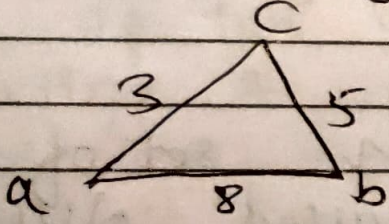
6) Directed Graph:-

A graph is said to be directed if every edge has direction. It is also called Di-graph.



7) Weighted Graph:-

A graph in which some way is assigned to every edge is called weighted graph.



8) Complete Graph: - (K_n)

A simple graph of n vertices in which degree of each vertex is $n-1$ and every vertex is connected to every other vertex, denoted by K_n

Properties :-

1) Every pair of vertices is adjacent in K_n .

2) Degree of each vertex in K_n is $n-1$.

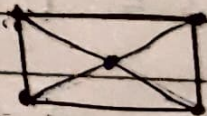
3) Total number of edges in K_n is $\frac{n(n-1)}{2}$.

⇒ Eg. K_3



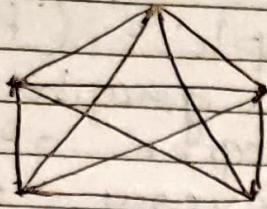
$$\begin{aligned} \text{deg of vertex} &= n-1 = 3-1 = 2 \\ \text{no. of edges} &= \frac{n(n-1)}{2} = 3 \end{aligned}$$

K_4



$$\begin{aligned} \text{deg of vertex} &= n-1 = 4-1 = 3 \\ \text{no. of edges} &= \frac{4(4-1)}{2} = 6 \end{aligned}$$

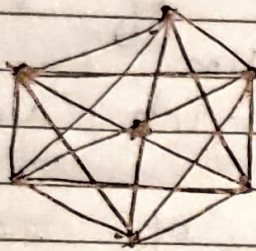
K_5



$$n-1 = 5-1 = 4$$

$$\frac{5(5-1)}{2} = \frac{20}{2} = 10$$

K_6



deg of vertex $\Rightarrow n-1 = 6-1 = 5$

$$\frac{6(6-1)}{2} = \frac{30}{2} = 15$$

Can a Complete Graph with 8 Vertices have 40 edges

$\Rightarrow$ For complete Graph with 8 vertices
no. of edges $= \frac{n(n-1)}{2} = \frac{8 \times 7}{2} = 28$

$\therefore K_8$ has 28 edges. Hence,
there does not exist a complete graph with
8 vertices & 40 edges

Regular Graph: 'r'

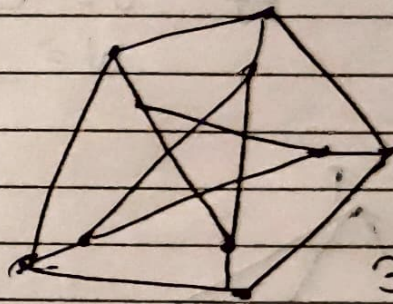
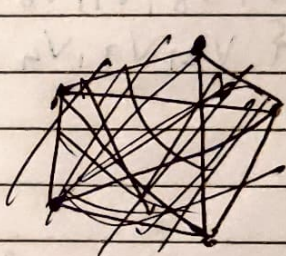
A graph in which every vertex has same degree, (say r). then it is called ' r ' regular graph

K_n is $n-1 \rightarrow$ regular graph.



3 regular graph
because each vertex
has 3 edges.

Note: - Every complete g is regular but converse is not true



Petersen graph

3-regular

1b) By Bipartite Graph

A graph $G = (V, E)$ is called

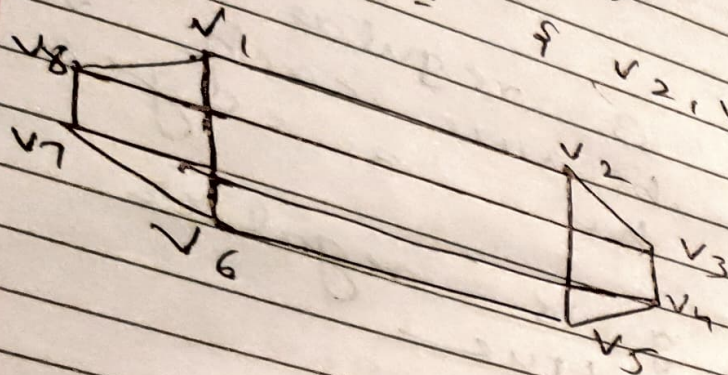
Bipartite, if

1) $V = U \cup W$ & $U \cap W = \emptyset$

2) Every edge E has 1 edge in V .

HW

g.



$$W = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$$

$$U = \{v_1\}$$

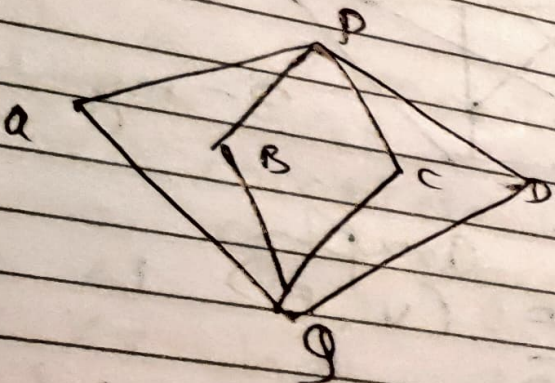
$$W = \{v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$$

$\Rightarrow$

$$U = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$$

$$W = \{v_2, v_3, v_4, v_5\}$$

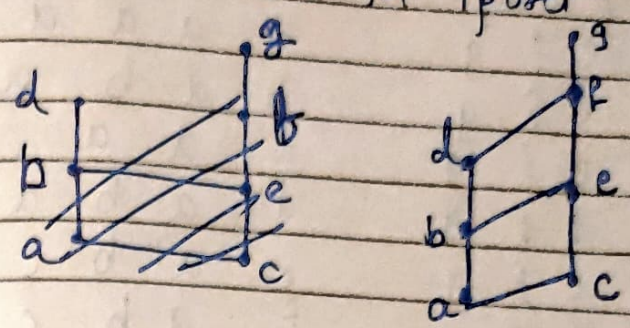
g



$$U = \{a, b, c, d\}$$

$$W = \{a, b, c, d\}$$

check whether $\{a, b, c, d, e, f, g\}$ is lattice or not.



Join -

v	a	b	c	d	e	f	g
a	a	b	c	d	e	f	g
b	b	b	e	d	e	f	g
c	e	e	c	g	e	f	g
d	d	d	f	d	f	f	g
e	e	e	e	f	e	f	g
f	f	f	f	f	f	f	g
g	g	g	g	g	g	g	g

Meet -

$\wedge$	a	b	c	d	e	f	g
a	a	a	a	a	a	a	a
b	a	b	a	b	b	b	b
c	a	a	c	a	c	c	c
d	a	b	a	d	b	d	d
e	a	b	c	b	e	e	e
f	a	b	c	d	e	f	f
g	a	b	c	d	e	f	g

Since $a \vee b, a \wedge b$ exist $\forall a, b \in L$.
implies, L is lattice

Complete Bipartite Graph:-

Let $G(V, E)$ is called complete bipartite graph if

i) G is bipartite.

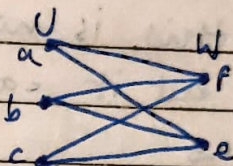
ii) G is simple graph

iii) $V = U \cup W$, as a partition of V .

Then every vertex in U is adjacent to every vertex in W . & if

U has m vertex, & W has n vertices then complete bipartite graph is denoted by $K_{m,n}$

Eg. $K_{3,2}$



- 6 edges, 5 vertices.

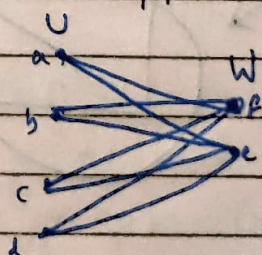
1) No. of vertices $K_{m,n} = m + n$

2) No. of edges in $K_{m,n} = m \times n$.

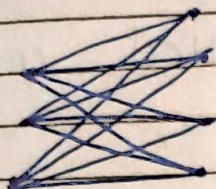
3) Degree of vertices in $W = n$

4) Degree of vertices in $U = m$

Eg. $K_{4,2}$



eg $K_{3,4}$



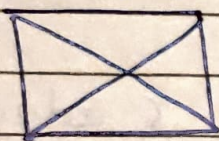
}

Planar graph.

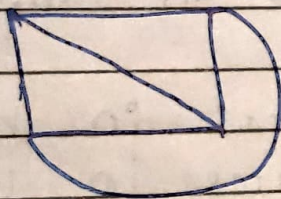
If it is possible to draw the diag of given graph in such a way that no two edges intersect, then it is called a planar graph.

eg.

①

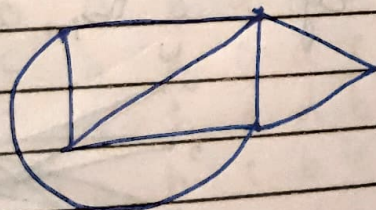
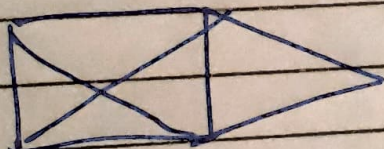


← this is not planar graph but it can be modified as-

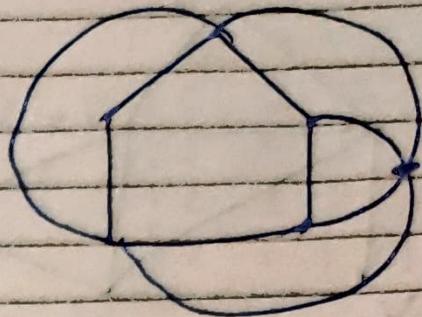
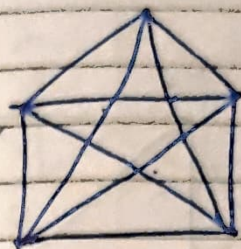


- planar graph.

②



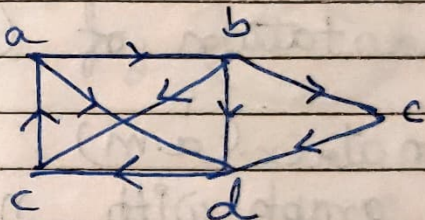
③



Not a planar

Indegree - In a directed graph, G , the no. of edges ending at vertex v is called the indegree of v . denoted by $d^+(v)$

Outdegree - In a directed graph G , the no. of edges beginning at vertex v is called outdegree of v . denoted by $d^-(v)$.



$$d^+(a) = 1$$

$$d^-(a) = 2$$

$$d^+(b) = 1$$

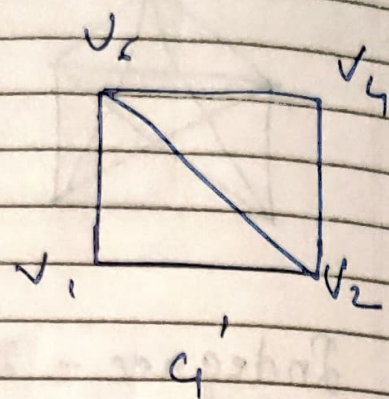
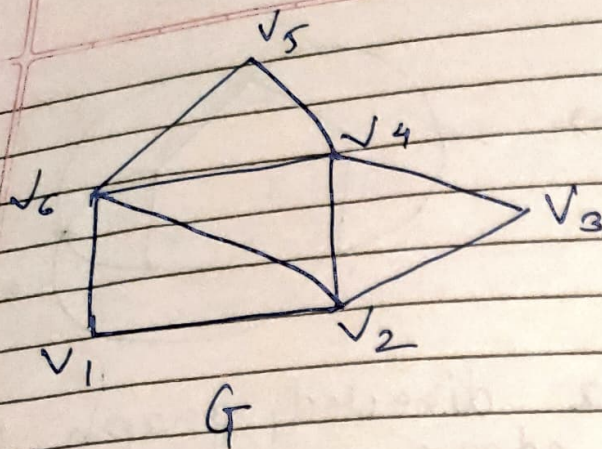
$$d^-(b) = 3$$

$$d^+(c) = 1 \quad d^+(d) = 3 \quad d^+(e) = 2$$

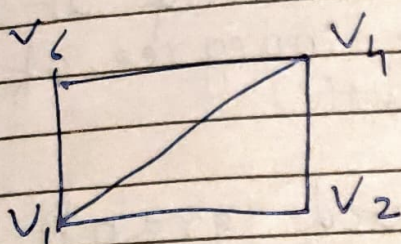
$$d^-(c) = 1 \quad d^-(d) = 1 \quad d^-(e) = 1$$

* Subgraph

A graph G' is called subgraph of a graph G , if all vertices of G' are in G & all the edges of G' are in G .



it is subgraph



It is not subgraph since edge given ~~before~~ $v_1, v_4 \notin G$.

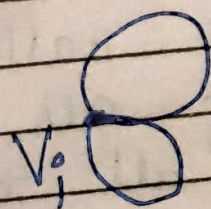
* Matrix Representation of Graph

1. Adjacency matrix (a.m)

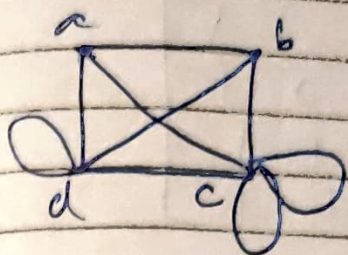
Let G be a graph with n vertices and no parallel edges then a.m of G is denoted as $A(G) = [a_{ij}]_{n \times n}$

$a_{ij} = 1$ if $v_i \neq v_j$ & adjacent

$a_{ij} = 0$, if $v_i \neq v_j$ are not adj.



if there is loop for vertex then count it as 1.

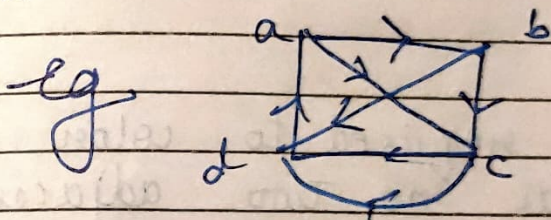


$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}_{4 \times 4}$$

a) ~~Adj~~ A.M of a digraph:-

Let G be a digraph with n vertices, the adjacency matrix of G denoted by $A(G) = [a_{ij}]_{n \times n}$ where $a_{ij} = 1$

$a_{ij} = 1$, if (v_i, v_j) belongs $(E) E$
 $a_{ij} = 0$, otherwise



$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 2 & 0 \end{bmatrix} \end{matrix}$$

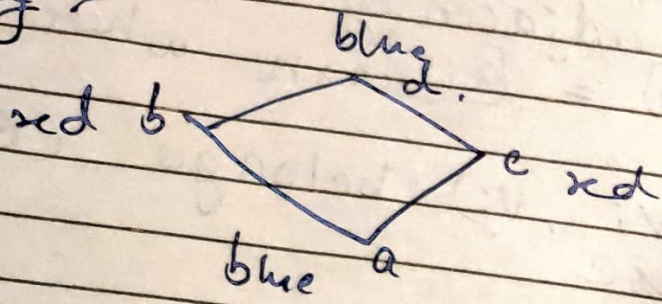
A.M for a graph is symmetric matrix
 but - A.M for a digraph need
 not be symmetric.

Coloring of a graph :-

adj vertices should have diff color.

Eg :-

①

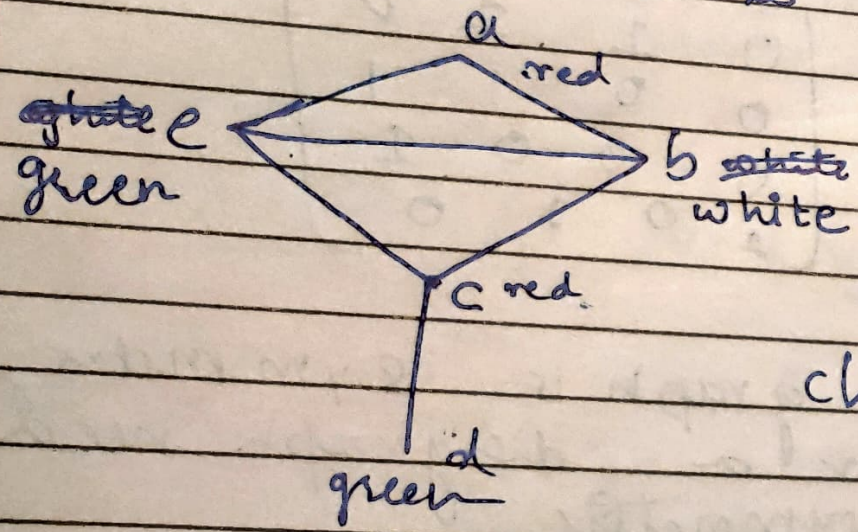


chromatic no $\rightarrow 2$.

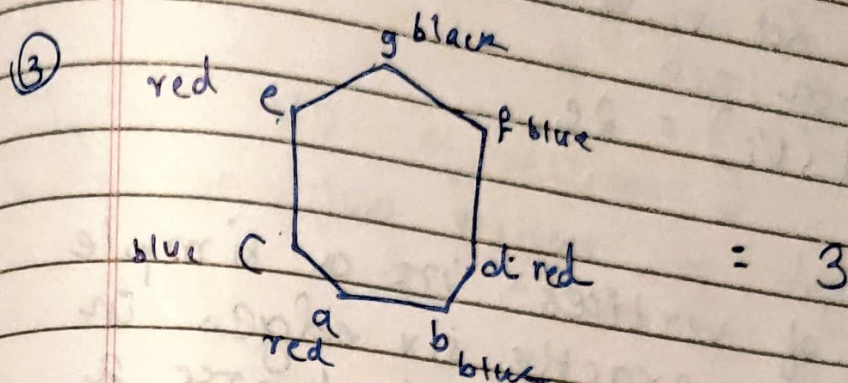
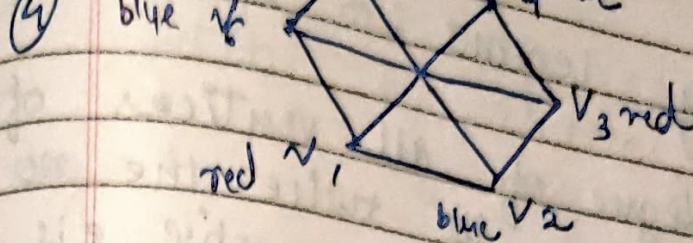
Explanation :-

Min. colors required to color the graph such that no two adjacent vertices receive same color. is 2.
 $\therefore$ chromatic no is 2 = $\chi(G) = 2$

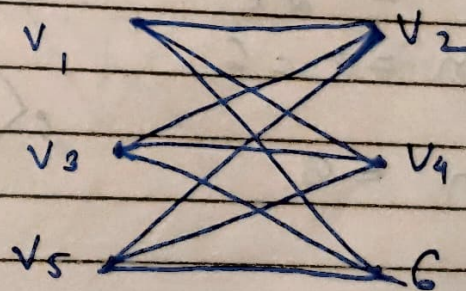
②



chromatic no. = $\chi(G) = 3$



1. A graph with one circuit and n vertices is two chromatic, if n is even.
2. It is 3 chromatic, if n is odd.
3. Any planar graph G can be colored using five or less color.
4. A complete graph K_n is n -chromatic graph.
5. Every two chromatic graph is bipartite. The G is also bipartite color. Collect coloring same color vertices in set.



* Hand shaking lemma :-

The sum of degree of all vertices of any graph is equal to twice the no. of edges. Let $v_1, v_2 \dots v_n$ are vertices & e is no. of edge then :-

$$\sum_{i=1}^n d(v_i) = 2e$$

Q. Find no. of vertices in a simple graph with exactly six edges in which each vertex is of degree 2. Draw such a graph.

No. of edges = $e = 6$.

degree of each vertex = 2

say no. of vertices = n

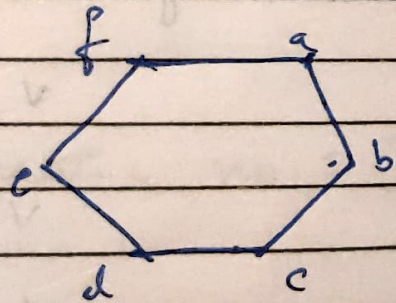
$$\sum_{i=1}^n d(v_i) = 2e.$$

$$2 + 2 + 2 + \dots + n = 2e$$

$$2n = 2e$$

$$n = e = 6$$

$$\therefore n = 6.$$



Q Find no. of edges in a graph with 6 vertices, two of degree 4 & 4 of degree 2.

2 deg 4
4 of deg 2.

No. of vertices = 6.

two vertices of deg-4
& 4 vertices of deg-2

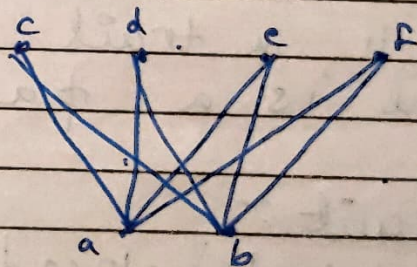
Let no. of edges = e .

By handshaking lemma

$$\sum_{i=1}^6 d(v_i) = 2e$$

$$4 + 4 + 2 + 2 + 2 + 2 = 2e$$

$$e = 8$$



Simple graph

* Trail :- a walk and edges can be repeated.

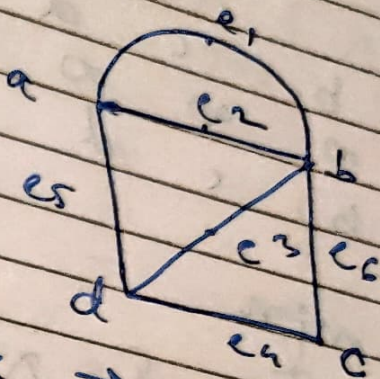
If edges of walk are distinct then such a walk is called a trail. On a trail, no edge is repeated.

If a vertices of a walk are distinct then such a walk is called a path.

* Every path is a trail, but not every trail is a path.

* Cycle / Circuit :-
A non-trivial closed trail (starting & ending vertex is same) in a graph is called a cycle or a circuit.

edge of circuit passes through every edge of graph.



- 1) Walk $\rightarrow$ $a e_1 b e_2 d$
 $a e_5 d e_3 b e_6 c e_4 a$
- 2) Trail $\rightarrow$ $d e_3 b e_6 c e_4 a$
- 3) Path $\rightarrow$ $a e_2 b e_3 d e_5 c$
- 4) closed path $\rightarrow$ $a e_2 b e_6 c e_4 d e_5 a$
- 5) cycle/circuit (closed trail) $\rightarrow$ $a e_1 b e_2 a$

6) Eulerian trail $\rightarrow$
 $a e_1 b e_2 a e_5 d e_3 b e_6 c e_4 d$

Step 1:-

	v	a	b	c	d	e	z
$\lambda(v)$	0	∞	∞	∞	∞	∞	∞
T	a	b	c	d	e	z	

$$\lambda(a) = 0$$

$$a \rightarrow \min_{b, c, d}$$

$$\lambda(b) = \min \{ \lambda(b), \lambda(a) + w(a, b) \}$$

$$= \min \{ \infty, 0 + 1 \} = 1$$

$$\lambda(c) = \min \{ \lambda(c), \lambda(a) + w(a, c) \}$$

$$= \min \{ \infty, 0 + 4 \} = 4$$

$$\lambda(d) = \min \{ \lambda(d), \lambda(a) + w(a, d) \}$$

$$= \min \{ \infty, 0 + 9 \} = 9$$

Step 2:-

	v	a	b	c	d	e	z
$\lambda(v)$	0	0	1	4	9	∞	∞
T	-	b	c	d	e	z	

$$\lambda(b) = 1$$

$$b \rightarrow \min_{c, d, e}$$

$$\lambda(c) = \min \{ \lambda(c), \lambda(b) + w(b,c) \}$$

$$= \min \{ 4, 1+2 \} = 3$$

$$\lambda(b) = \min \{ \lambda(d), \lambda(b) + w(b,d) \}$$

$$= \min \{ 9, 1+7 \} = 8$$

$$\lambda(e) = \min \{ \lambda(e), \lambda(b) + w(b,e) \}$$

$$= \min \{ \infty, 1+5 \} = 6$$

v	a	b	c	d	e	z
$\lambda(v)$	0	1	3	8	6	∞
T	-	-	c	d	e	z

$$P = \lambda(c) = 3 \text{ minimum}$$

$$c \rightarrow z, e$$

$$\lambda(z) = \lambda(z) = \min \{ \lambda(z), \lambda(c) + w(c,z) \}$$

$$d \rightarrow \cdot \quad \min \{ \infty, 3+9 \} = 12$$

$$\lambda(e) = (\lambda(e), \lambda(c) \min + w(c,e) = \{ 6, 3+1 \}$$

$$d \rightarrow 4$$

Step 4:-

v	a	b	c	d	e	z
$\lambda(v)$	0	1	3	8	4	12
T	-	-	-	d	e	z

$$\lambda(e) = 4 \text{ min}$$

$$e \rightarrow d, z$$

$$\lambda(d) = \min \{ \lambda(d), \lambda(e) + w(e,d) \}$$

$$= \min \{ 8, 4+3 \} = 7$$

$$\lambda(z) = \min(\lambda(z), \lambda(e) + w(z, e))$$

$$= \min\{12, 4 + 6\} = 10$$

Step 3:-

v	a	b	c	d	e	z
$\lambda(v)$	0	1	3	7	4	10
T	-	-	-	d	-	z

$$\lambda(d) = 7 \min$$

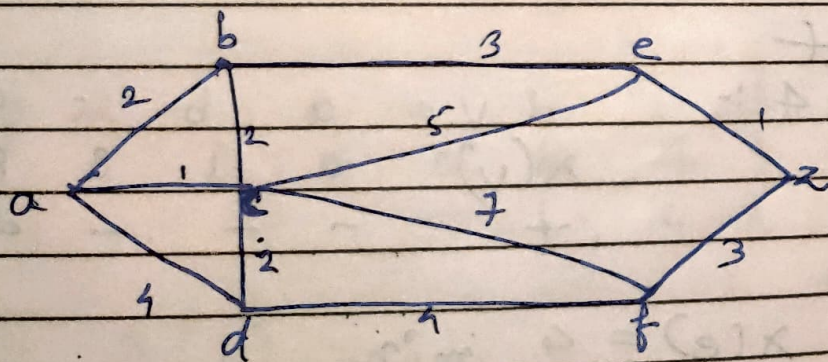
$$d \rightarrow z$$

$$\lambda(z) = \min(\lambda(z), \lambda(d) + w(d, z))$$

$$= \min\{10, 7 + 2\} = 9$$

v	a	b	c	d	e	z
$\lambda(v)$	0	1	3	7	4	9
T	-	-	-	-	-	z

The shortest path from a to z is, diagram
 & the shortest distance is $1 + 2 + 1 + 3 + 2 = 9$.



WRONG QUESTⁿ

Step 1:-

v	a	b	c	d	e	f	z
$\lambda(v)$	0	∞	∞	∞	∞	∞	∞
T	a	b	c	d	e	f	z

$$\lambda(a) = 0 \text{ min}$$

a $\rightarrow$ b, c, d

$$\lambda(b) = \min(\lambda(b), \lambda(a) + w(a, b))$$

$$\min(\infty, 0 + 2) = 2$$

$$\lambda(c) = \min(\lambda(c), \lambda(a) + w(a, c))$$

$$\min(\infty, 0 + 1) = 1$$

$$\lambda(d) = \min(\lambda(d), \lambda(a) + w(a, d))$$

$$\min(\infty, 0 + 4) = 4$$

Step 2:-

v	a	b	c	d	e	f	z
$\lambda(v)$	0	2	1	4	∞	∞	∞
T	-	b	c	d	e	f	z

$$\lambda(c) = 1 \text{ min}$$

c $\rightarrow$ b, d, e, f

$$\lambda(b) = \min\{2, 1 + 2\} = 2$$

$$\lambda(d) = \min\{4, 1 + 2\} = 3$$

$$\lambda(e) = \min\{\infty, 1 + 5\} = 6$$

$$\lambda(f) = \min\{\infty, 1 + 7\} = 8$$

Step 3:-

v	a	b	c	d	e	f	z
$\lambda(v)$	0	2	1	3	6	8	∞
T	-	b	-	d	c	f	z

$$\lambda(b) = 2 \text{ min}$$

$b \rightarrow \text{X, X, e.}$

$$\lambda(e) = \min \{ 6, 2+3 \} = 5$$

Step 4:-

v	a	b	c	d	e	f	z
$\lambda(v)$	0	2	1	3	5	8	∞
T	-	-	-	d	e	f	z

$$\lambda(d) = 3 \text{ min}$$

$d \rightarrow f.$

$$\lambda(f) = \min \{ 8, 3+4 \} = 7.$$

Step 5:-

v	a	b	c	d	e	f	z
$\lambda(v)$	0	2	1	3	5	7	∞
T	-	-	-	-	e	f	z

$$\lambda(e) = 5 \text{ min}$$

$e \rightarrow z$

$$\lambda(z) = \min \{ \infty, 5+1 \} = 6$$

Step 6:-

V	a	b	c	d	e	f	z
$\chi(v)$	0	2	1	3	5	7	6
T	-	-	-	-	-	F	Z

$$\chi(z) = 6 \text{ min}$$

Z -

Note:-

Values are wrong, so quesⁿ is wrong

* H.W:-

